Layout-Preserving Scientific Translation

TransYH Development Team | Synthetic Evaluation Paper

Abstract. Scientific papers contain text, mathematical equations, images, and tables. This synthetic document evaluates whether a local PDF translation system can preserve the geometry of these elements while replacing English prose with Chinese. It is an engineering fixture, and its numerical values are illustrative.

1. Introduction

Researchers need to read scientific literature in their preferred language. A useful translation workflow should preserve the original document and save the translation as a separate attachment. Readers should switch between documents in one tab without losing their reading position or changing the zoom level.

2. Experimental Setting

We use an original three-page document with a fixed page size. The first page contains ordinary English prose. The second page combines a mathematical equation, an embedded image, and a table. The final page uses two columns. All text and graphics were created specifically for this test.

Mathematical Model and Visual Evidence

The following equation describes a simple relationship between mass and energy. Mathematical symbols should remain unchanged during translation. The figure and table are placed at fixed coordinates to assess layout preservation.

$$E = mc^2 \tag{1}$$

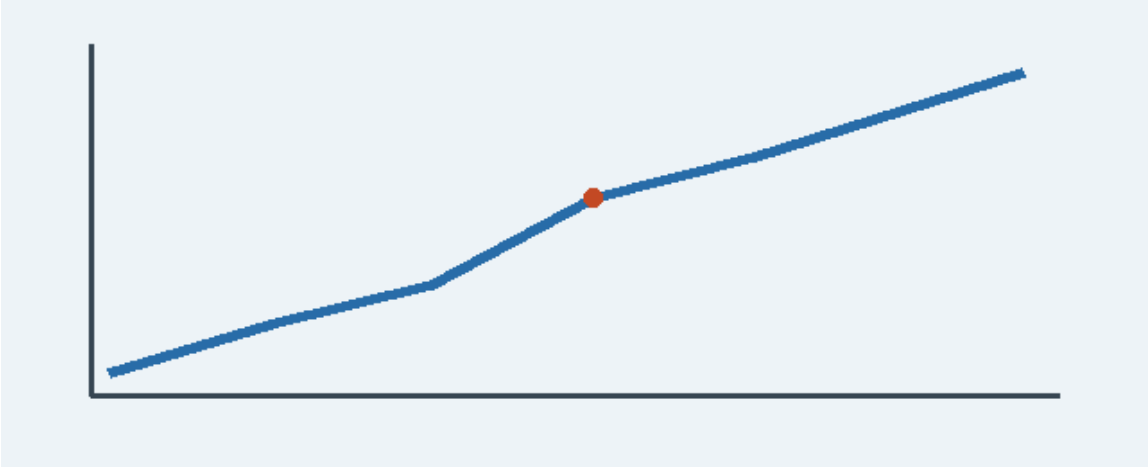

Figure 1. A synthetic trend used only for image preservation testing.

Table 1. Illustrative measurements

Method	Precision	Recall
Baseline	0.81	0.78
Proposed	0.87	0.84

Two-Column Results and Discussion

The translation system must treat the two columns as separate reading regions. Each translated paragraph should fit its original region without covering the adjacent column. Keeping the page dimensions and page count unchanged allows reliable navigation between the original paper and its translation. This experiment measures software behavior and does not establish the linguistic quality of a real language model.

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